Phyloseq Visualizations in Galaxy

BIOL 4810

2026-04-16

Goal: Use **phyloseq in Galaxy** to visualize your microbial community data with **alpha diversity**, **beta diversity**, and a **taxonomic bar plot**.
Submission: Fill in answers, upload screenshots where prompted, then click **Prepare submission preview** and **Download as PDF (Print)**.

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0) Your dataset

0A. Sample set / project name:

WCSP Microbiome

0B. Grouping variable used for comparisons:

Sex

0C. Other metadata variable(s) available for plotting or faceting:

Infection Status

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phyloseq workflow focus

For this module, you will generate and interpret:

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Alpha diversity → Beta diversity (NMDS using Bray–Curtis) → Taxonomic
bar plot
```

Reminder: These visualizations help you describe diversity within samples, differences among samples, and which taxa are driving those patterns.

Part 1 — Alpha diversity

1A) Alpha diversity

In your own words, what is alpha diversity?

Alpha diversity is the measurement of the amount of microbial communities within a sample.

1B) Observed richness

What does Observed measure?

Observed richness is the amount of different types of bacteria found.

1C) Shannon diversity

What does Shannon measure?

Shannon measures how "mixed" the group is.

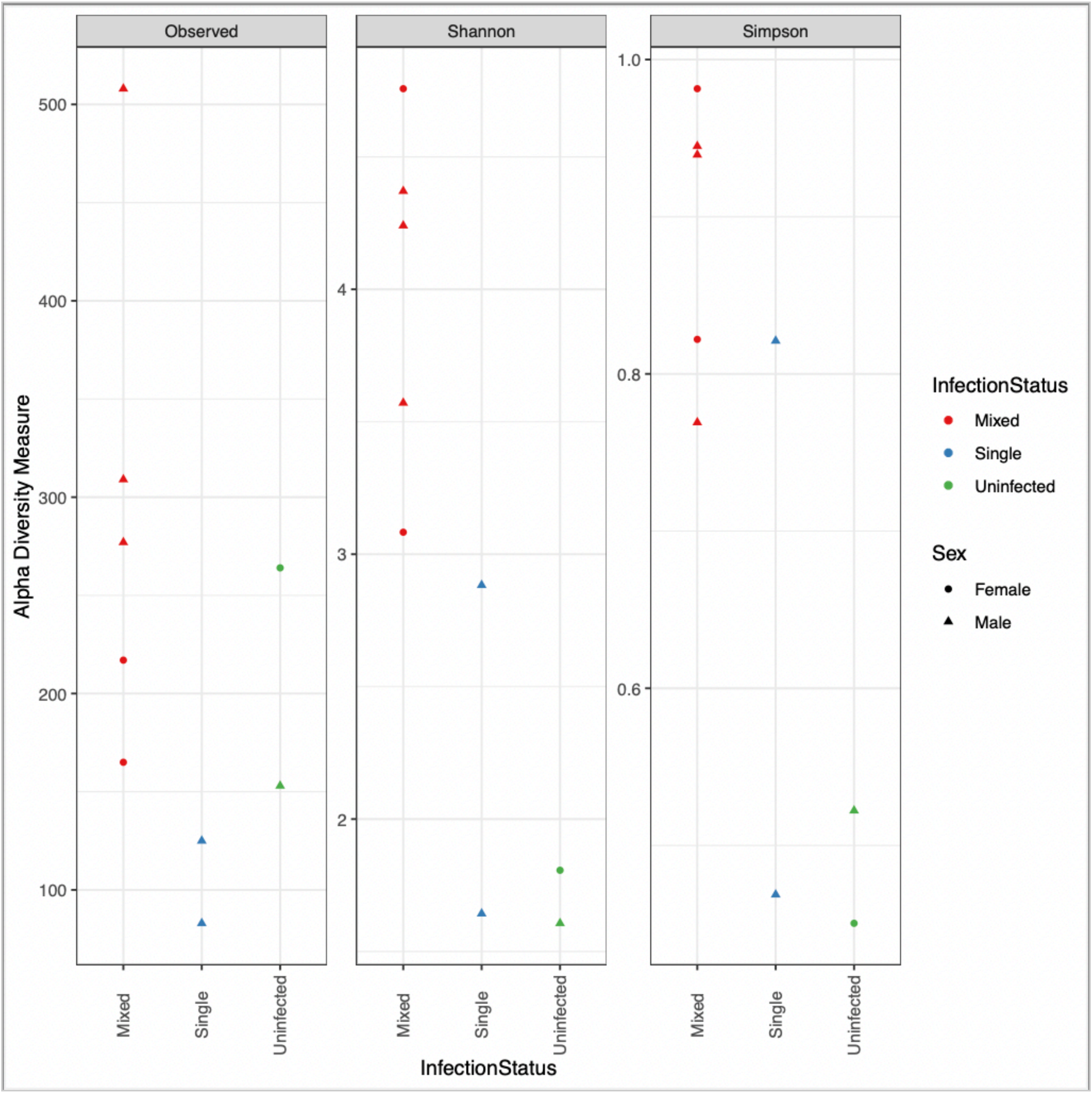
1D) Simpson diversity

What does Simpson measure?

Simpson measures the representation of each group.

Upload screenshot: alpha diversity plot

Choose File Screen Shot… 5.56.40 PM



Grouping variable shown on the alpha diversity plot:

Infection Status

Which sample or group appears to have the greatest alpha diversity?

Mixed

Did Observed, Shannon, and Simpson show the same pattern?

For the most part, yes. Usually if one had a group low, then the other did as well, same for those that were higher on the y-axis.

Part 2 — Beta diversity (NMDS with Bray-Curtis)

What beta diversity means:
Beta diversity describes differences in community composition **between samples**.

In your own words, what is beta diversity?

Beta diversity is the comparison of the diversity between samples.

2A) Bray-Curtis dissimilarity

What does Bray-Curtis measure?

Bray-Curtis is when the amount of bacteria from a sample is being compared; like, the amount of carrots used in a carrot cake recipe.

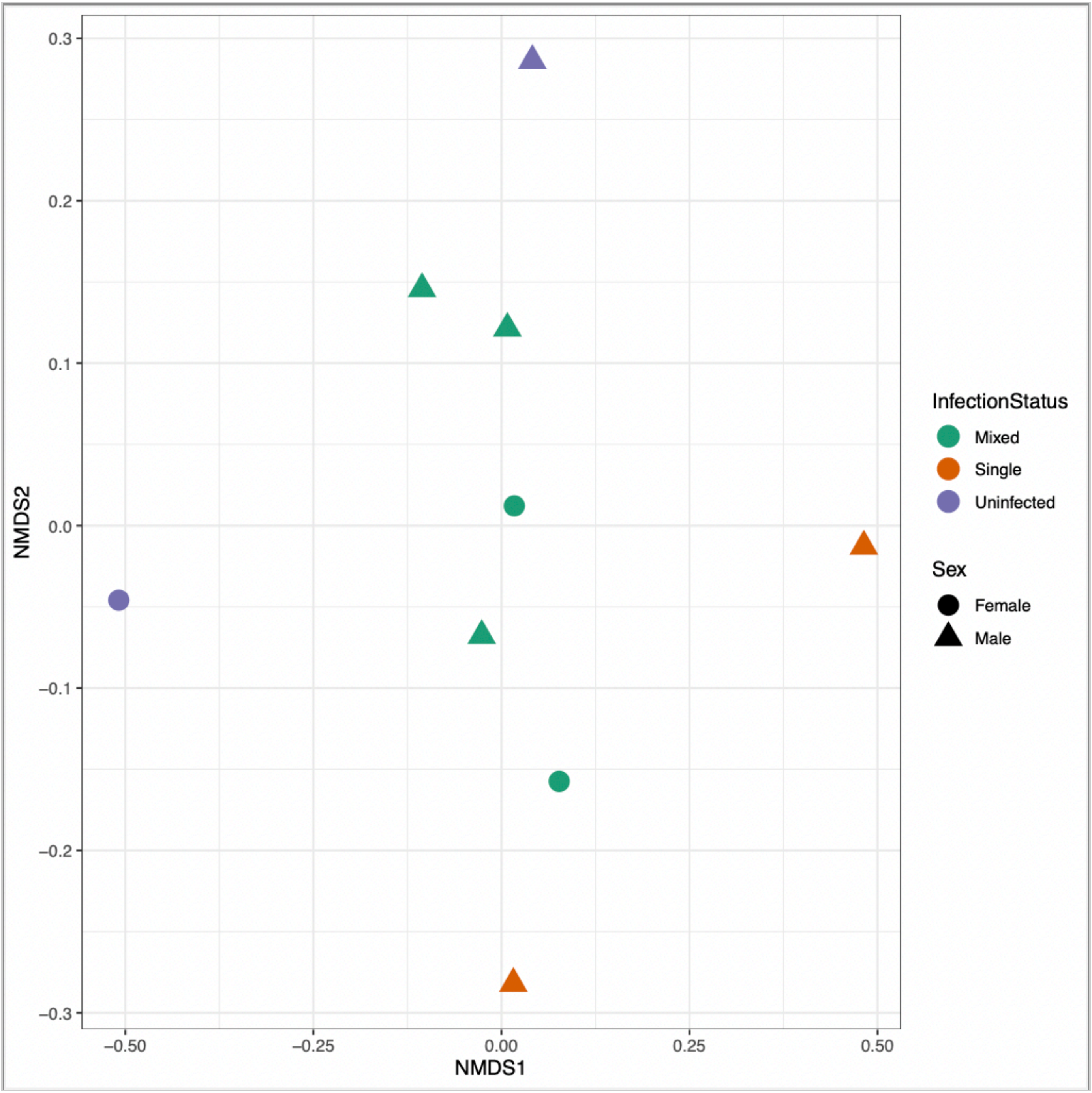
2B) NMDS

What does NMDS show?

NMDS explains similarity between things using distance.

Upload screenshot: NMDS Bray-Curtis plot

Choose File Screen Shot… 6.10.58 PM



Color variable used on the NMDS plot:

Infection Status

Shape variable used on the NMDS plot (if any):

Sex

Which samples or groups cluster together?

Mixed Males and Females

Are any groups clearly separated from one another? What might that suggest?

The uninfected and single groups are separated from each other by each one being on one end of the plot. This could mean that they're furthest in similarity

If points overlap strongly, what does that mean?

They're very similar

Part 3 — Taxonomic bar plot

What a taxonomic bar plot shows:
A bar plot summarizes the composition of samples by showing the relative abundance of taxa at a selected taxonomic level.

What taxonomic level did you use?

phylum

How did you facet the plot?

Infection Status

Why was this taxonomic level useful?

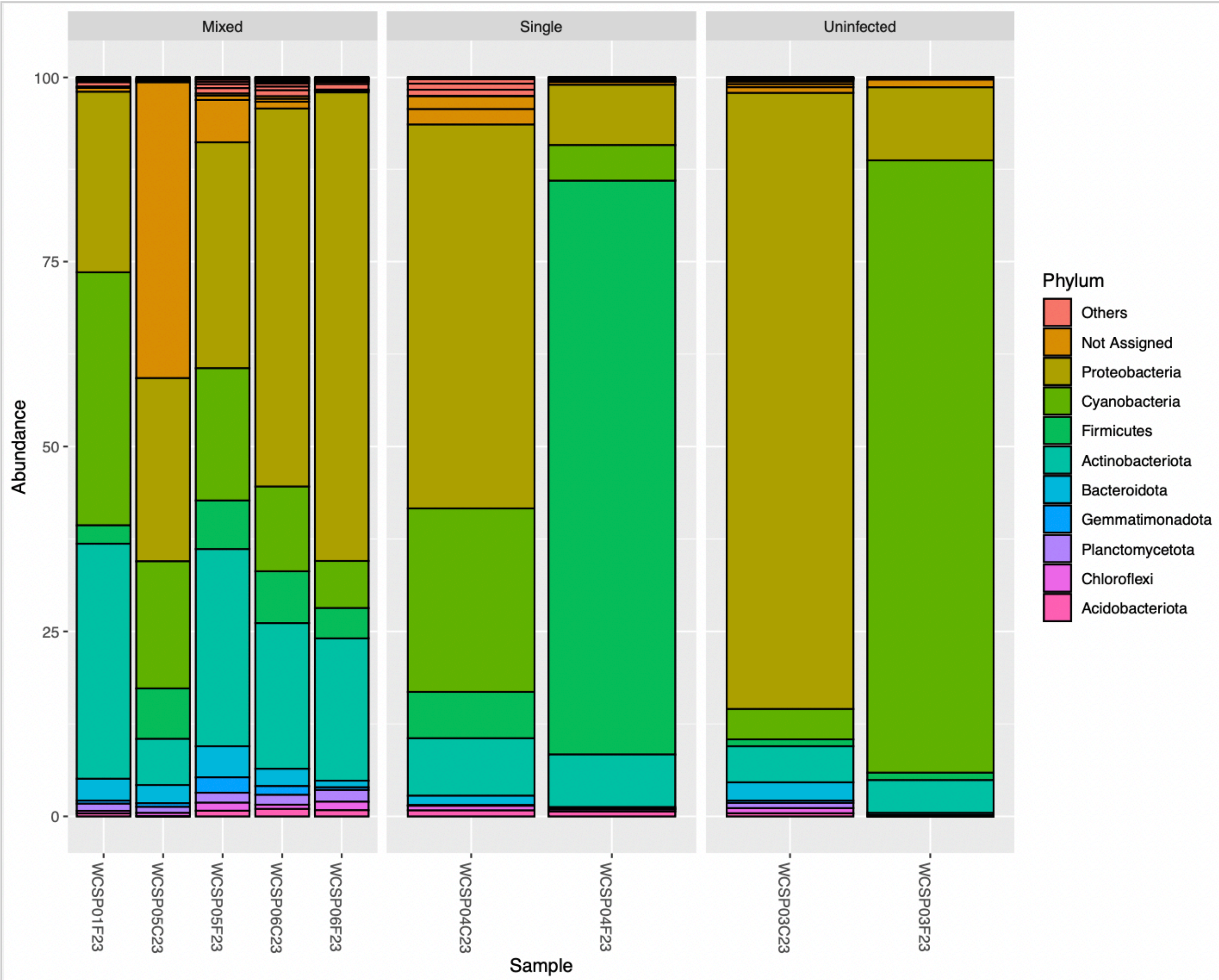
It allows us to look at the big picture of our data.

Why was your faceting choice useful?

It lets us compare the info we get.

Upload screenshot: taxonomic bar plot

Choose File Screen Shot… 6.18.29 PM



Which taxa appear to be most abundant?

Proteobacteria

Did taxonomic composition differ among groups?

For the most part, not very much.

Part 4 — Overall interpretation

In 3–5 sentences, summarize what your alpha diversity, beta diversity, and bar plot together suggest about this dataset.

My data is all based on the infection status of males & females. My alpha diversity and beta diversity suggest that the mixed samples have way more amounts of bacteria, while the beta samples suggest that the three groups are pretty distant in the types of bacteria they have. Then the bar plot tells us although the types of bacteria there is similar, the amount of the bacteria differs.

Export (PDF)

1. Click **Prepare submission preview** (generates a clean summary).
2. Click **Download as PDF (Print)** and save.

Submission preview

Dataset
Project/sample set: WCSP Microbiome
Grouping variable: Sex
Other metadata variables: Infection Status

Alpha diversity
Definition: Alpha diversity is the measurement of the amount of microbial communities within a sample.
Observed: Observed richness is the amount of different types of bacteria found.
Shannon: Shannon measures how "mixed" the group is.
Simpson: Simpson measures the representation of each group.
Grouping variable: Infection Status
Highest alpha diversity: Mixed
Did the metrics show the same pattern? For the most part, yes. Usually if one had a group low, then the other did as well, same for those that were higher on the y-axis.

Beta diversity (NMDS Bray-Curtis)
Definition: Beta diversity is the comparison of the diversity between samples.
Bray-Curtis: Bray-Curtis is when the amount of bacteria from a sample is being compared; like, the amount of carrots used in a carrot cake recipe.
NMDS: NMDS explains similarity between things using distance.
Color variable: Infection Status
Shape variable: Sex
Which groups cluster together? Mixed Males and Females
Are any groups separated? The uninfected and single groups are separated from each other by each one being on one end of the plot. This could mean that they're furthest in similarity
If points overlap strongly: They're very similar

Taxonomic bar plot
Taxonomic level: phylum
Faceted by: Infection Status
Why this taxonomic level? It allows us to look at the big picture of our data.
Why this faceting choice? It lets us compare the info we get.
Most abundant taxa: Proteobacteria
Did composition differ among groups? For the most part, not very much.

Overall interpretation
My data is all based on the infection status of males & females. My alpha diversity and beta diversity suggest that the mixed samples have way more amounts of bacteria, while the beta samples suggest that the three groups are pretty distant in the types of bacteria they have. Then the bar plot tells us although the types of bacteria there is similar, the amount of the bacteria differs.